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Averages vs. Extremes: Rethinking Climate Health and Weather

Abstract

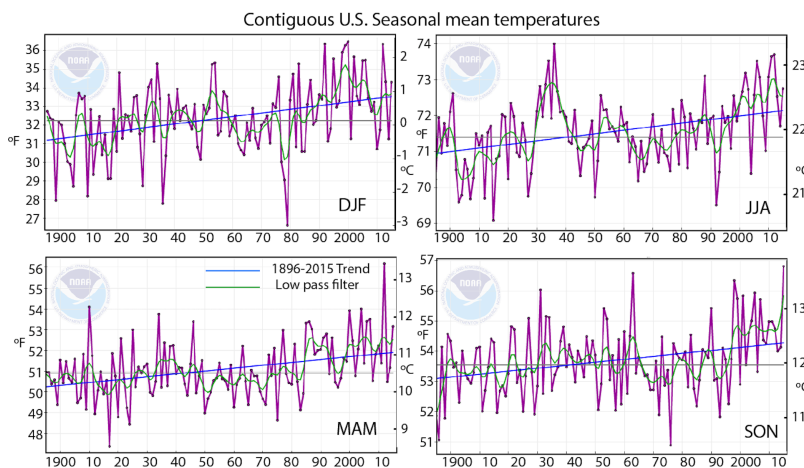
In 2016, a study in *Nature* argued that weather had become more pleasant for 80% of Americans over four decades, because winters grew milder while summers did not get much worse. Climate scientist Kevin Trenberth disagreed. He argued that looking at averages hides what matters most for health and safety: extremes such as heat waves, droughts, wildfires, floods, and storm surges.

Introduction

Researchers Patrick Egan and Megan Mullin reported that 80% of Americans live in counties with more pleasant weather than 40 years ago. They suggested this might explain why there had been little public demand for climate policy. But their analysis used only average temperatures in January and July and total yearly precipitation. As Trenberth pointed out, people do not live in averages. No one experiences long-term average weather, but we increasingly experience extremes and their impacts on our health, safety, and well-being. At the National Center for Atmospheric Research, he and his colleagues have studied extremes precisely because they destroy crops, wreck infrastructure, and threaten lives and property.

How Do Scientists Study This?

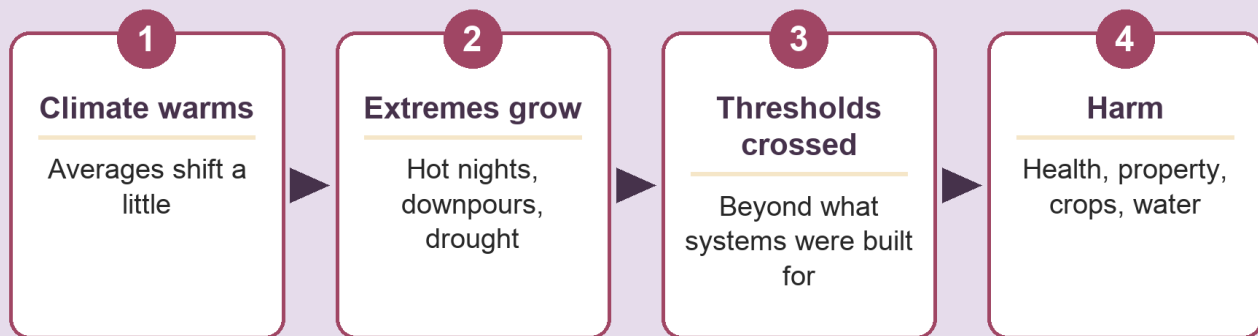
Trenberth evaluated the Nature study by examining what it measured and what it left out. He compared its focus on January and July averages with NOAA data on temperatures in all four seasons across the contiguous United States. He then reviewed evidence on extremes: the share of U.S. land with unusually hot summer nights, heavy downpour trends by region, California's snowpack and drought, wildfire seasons, insect outbreaks, and storm surges. His argument rests on a key principle: disasters often strike when a threshold is crossed, and our cities, farms, and infrastructure are built for the weather of the past.



FROM THE ARTICLE

Contiguous U.S. seasonal mean temperatures, 1896-2015: warming appears in all seasons, not just January and July. Adapted from NOAA data, author provided.

Why extremes matter more than averages



Study the Process

Check the measures

The study used only January and July averages and yearly rain totals.

Look at all seasons

NOAA data show warming in every season, with records outside January and July.

Examine extremes

Heat waves, downpours, drought, and storm surges reveal the real impacts.

What Did They Discover?

The share of U.S. land with unusually hot summer nights rose from under 10% in the 1970s to over 40% in recent years, giving less overnight relief to the young, elderly, and ill. The amount of rain falling in the heaviest downpours increased 20% nationally over the past century, 71% in the Northeast and 37% in the Midwest, raising flood risk. In California, the Sierra Nevada snowpack, which supplies nearly 30% of the state's water, hit its lowest level in at least 500 years. Warm winters let pine beetles survive, contributing to the death of 46 million acres of trees from 2000 to 2012. And rising seas make storm surges like Superstorm Sandy's far more destructive.

Extremes by the numbers

10%→40%

U.S. land with unusually hot summer nights

71%

more rain in the heaviest Northeast downpours

46M

acres of trees killed as pine beetles survived warm winters

Adding climate change to natural variability can become the straw that breaks the camel's back, pushing extremes past the thresholds our infrastructure was built for.

Look at the Evidence

FROM THE ARTICLE



The 2015 wildfire season was the largest in Washington state history. Image: MODIS.



The Jersey shore after Superstorm Sandy. Photo: Anthony Quintano, CC BY-SA.

Takeaways

Averages hide danger

Pleasant averages can mask more dangerous heat, floods, and drought.

Extremes cause harm

Most loss of life and damage happens when records are broken.

Built for the past

Cities and farms designed for old weather are vulnerable to new extremes.

Key Terms

Weather extreme

An unusually severe weather event, such as a record heat wave, torrential rain, or drought.

Threshold

A tipping point beyond which conditions cause damage, such as a flood level that overtops a levee.

Snowpack

Accumulated snow in mountains that melts in spring and summer, supplying water to rivers and reservoirs.

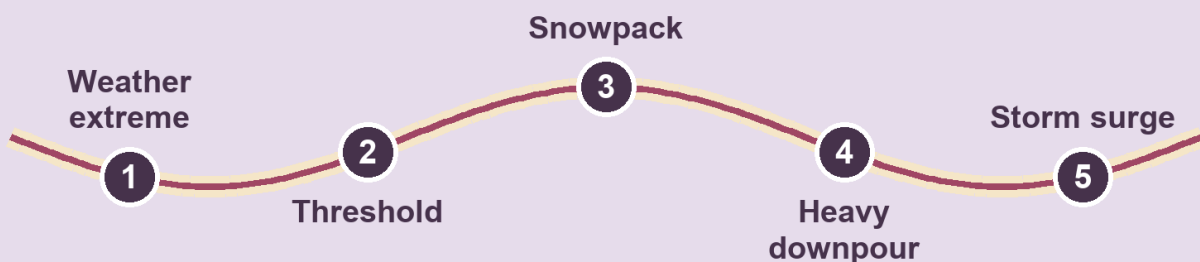
Heavy downpour

Intense rainfall in a short time, which can overwhelm drains and cause flooding.

Storm surge

A rise in sea level caused by a storm's winds pushing water onshore, made worse by higher sea levels.

Vocabulary Path · Averages vs. Extremes



Activity

Average or Extreme?

Students discover how the same average can hide very different extremes, and why extremes matter for health.

Materials: Two made-up week-long temperature records with the same average; Calculator; Graph paper

1. Calculate the average temperature for each week. Notice they are the same.
2. Graph both weeks. Describe how they differ.
3. Identify which week would be more dangerous for an older adult without air conditioning, and why.
4. Connect to the lesson: find two statistics that describe extremes rather than averages.
5. Write a response to the claim that U.S. weather has become more pleasant.

References

Trenberth, K. (2016). Has climate change really improved U.S. weather? The Conversation. <https://doi.org/10.64628/AAI.sx45kwajh>

Egan, P. J., and Mullin, M. (2016). Nature, April 20, 2016, as described in the article above.